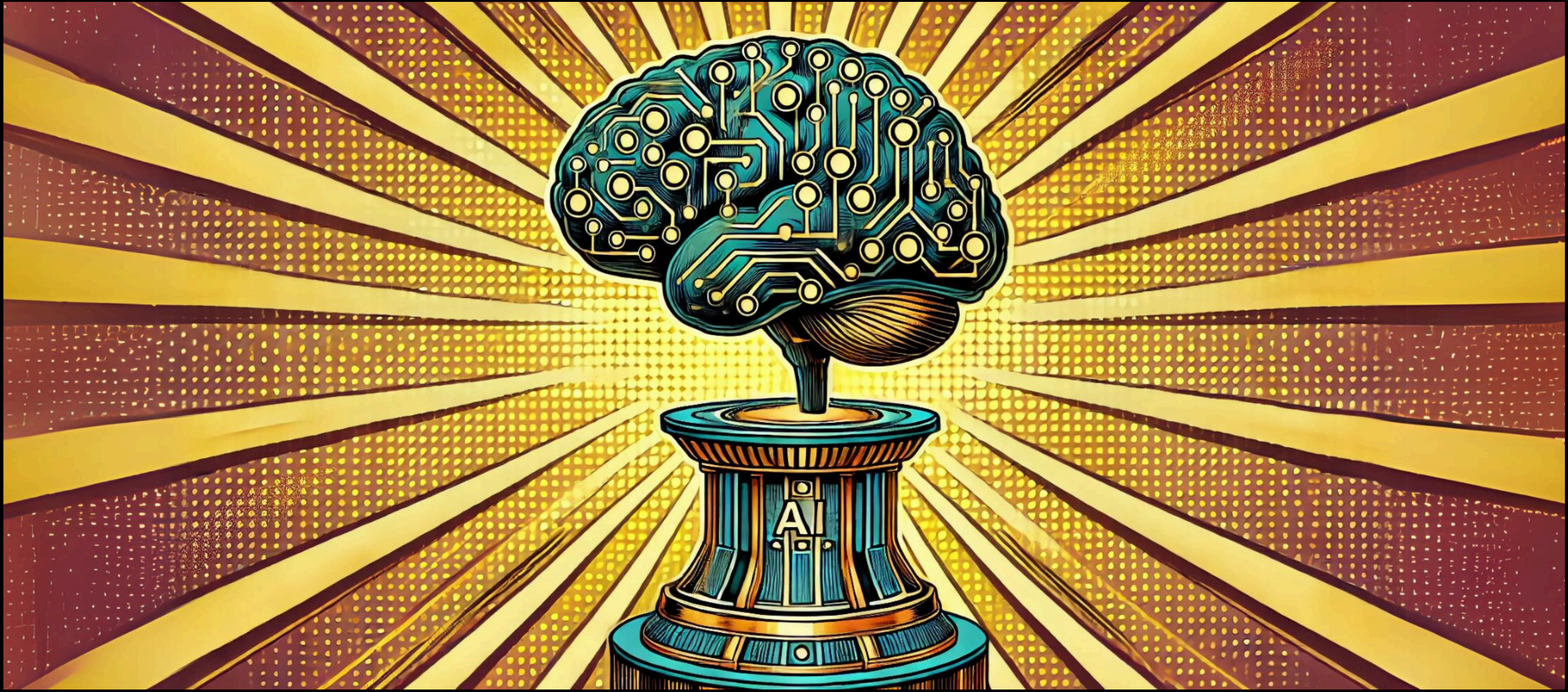
LLM Engineering

MASTER AI & LARGE LANGUAGE MODELS



WEEK 8

GAME **ON**

What you can now do

- Generate text and code with Frontier Models and Open Source models using APIs and HuggingFace, including tools, assistants and RAG
- Follow a 5 step strategy to solve problems, including dataset curation, making a baseline model, and fine-tuning a Frontier model
- Confidently carry out the end-to-end process for selecting and training specialized open source models that can outperform the Frontier

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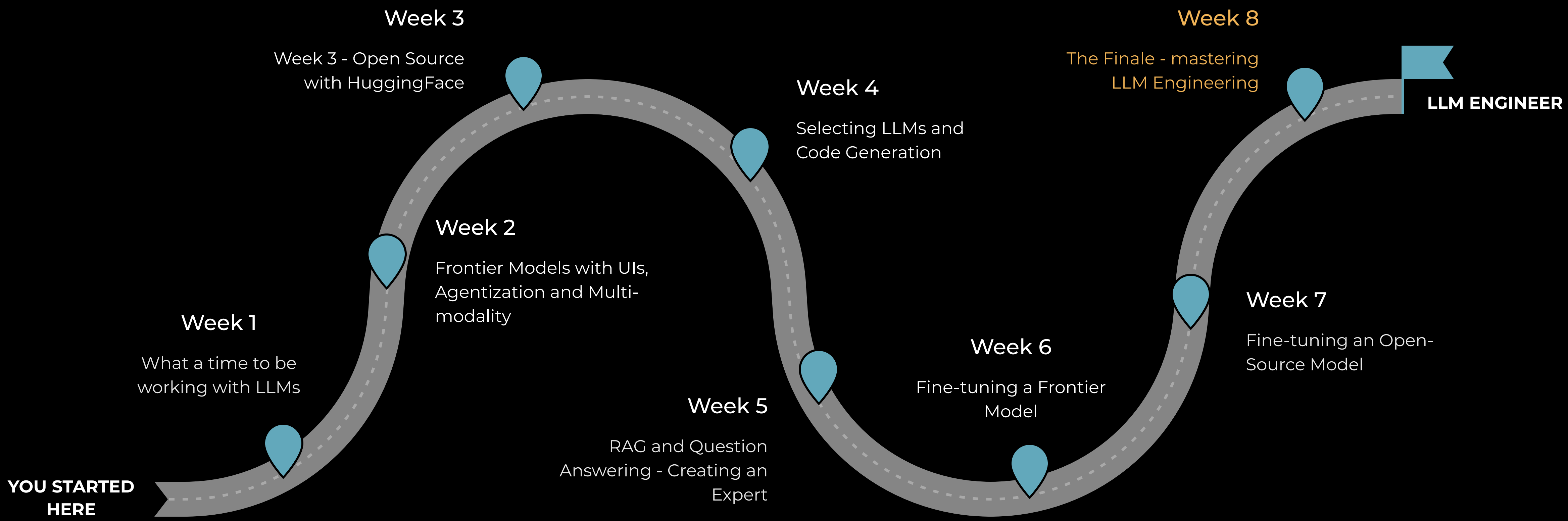
An intensive week lies ahead! After today, you'll be able to:

- Use Modal, the serverless platform for AI, to run code remotely
- Deploy an LLM behind an API in the cloud
- Create an Agent that will be incorporated into an autonomous Agentic AI solution

This week the training wheels are OFF



It's all been building up to this

Capstone Project - "**The Price Is Right**"

We will build an Autonomous Agentic AI framework

- Watches for deals being published online
- Estimates the price of the product
- Sends a push notification when it finds great opportunities

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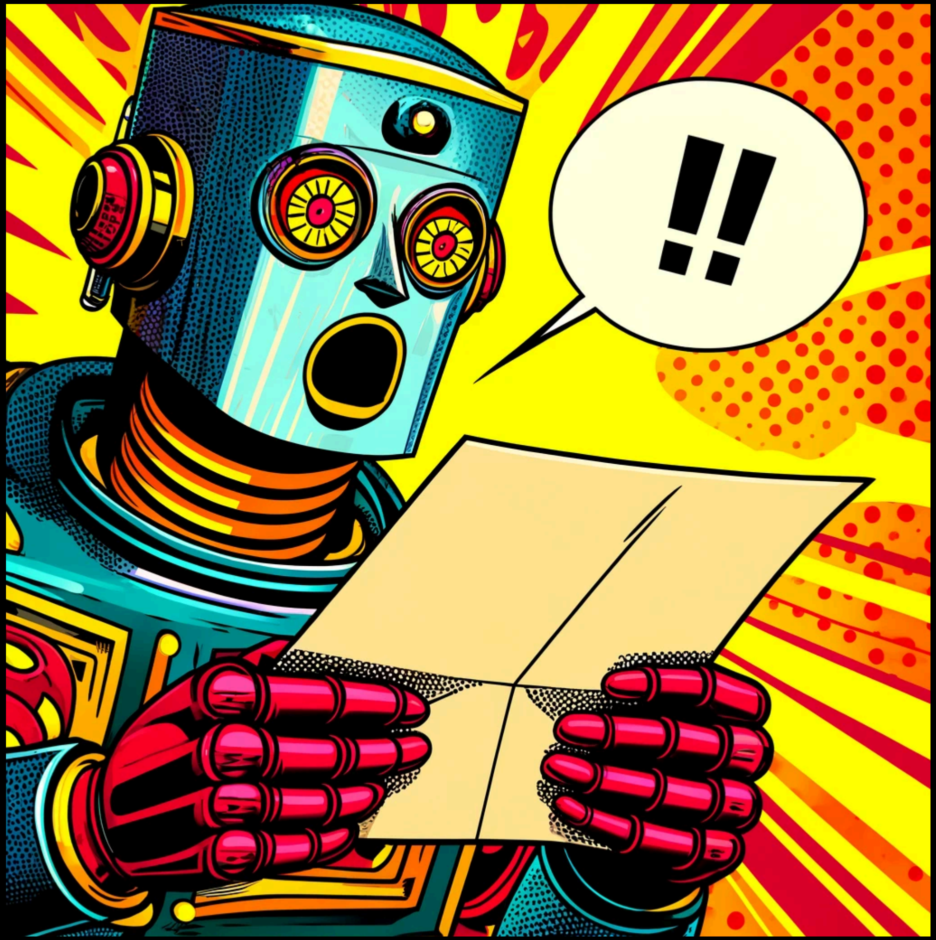
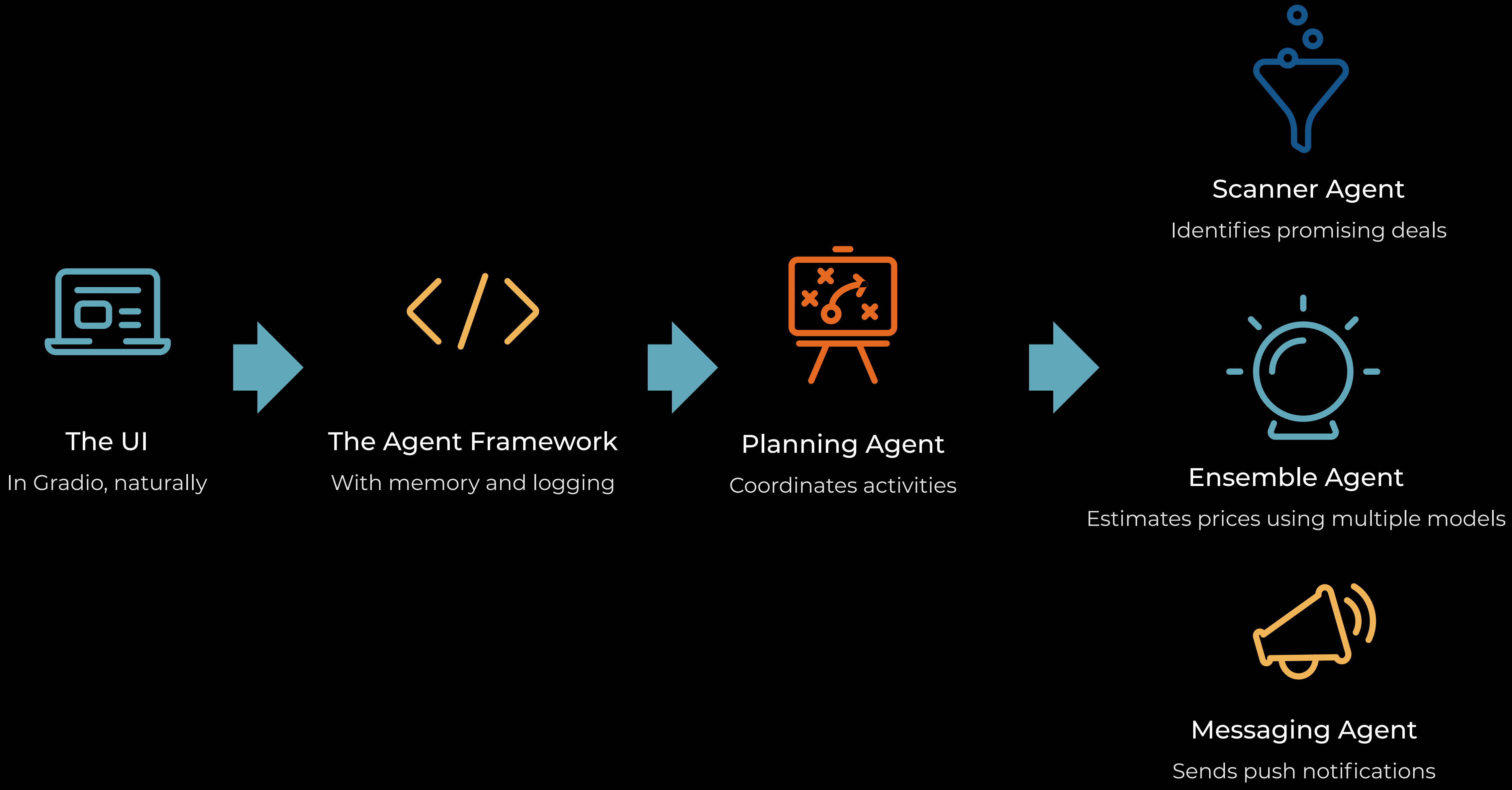
SEVEN Agents will collaborate in our framework

- GPT-4o model will identify the deals from an RSS feed
- Our frontier-busting fine-tuned model will estimate prices
- We'll also use a Frontier Model with a massive RAG db



Agent Architecture



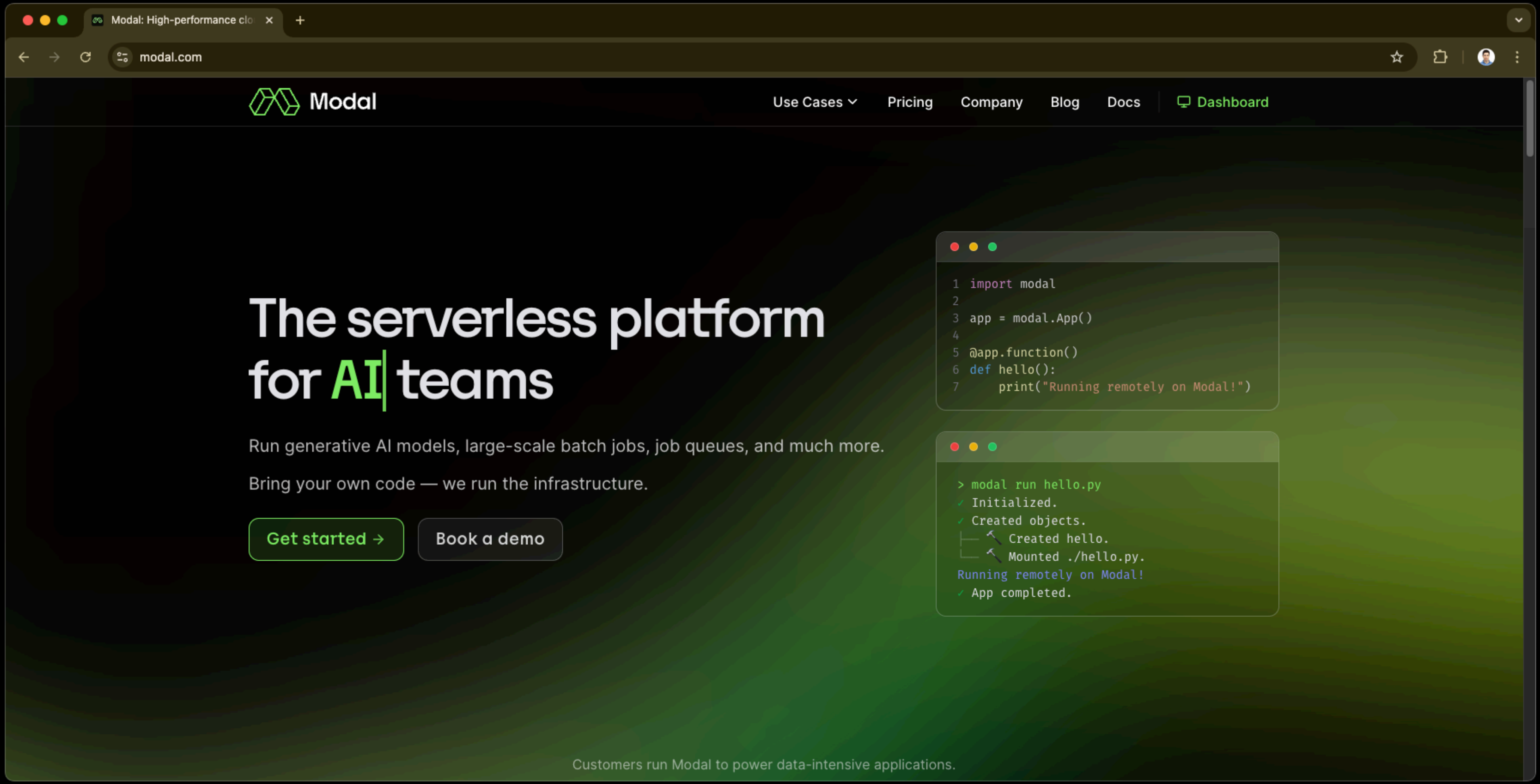
As we head towards productionizing

Type hints

Comments

Logging

Introducing Modal



WEEK 8

I told you this week would be big!

What you can now do

- Generate text and code with Frontier Models and Open Source models using APIs and HuggingFace, including tools, assistants and RAG
- Follow a 5 step strategy to solve problems, including dataset curation, making a baseline model, and fine-tuning a Frontier model
- Confidently carry out the end-to-end process for selecting and training specialized open source models that can outperform the Frontier
- Use Modal to deploy LLMs for production

By end of next time you'll be able to:

- Confidently build an advanced RAG solution, without Langchain
- Build an ensemble model with a high level of expertise
- Deliver production ready code to call a number of models